LAB TITLE:- IOS(iPhone SE iOS 13.4.1) FORENSIC

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STUDENT ID:- 2025FWSD11563

COURSE NAME:- MOBILE AND IOT FORENSICS CASE STUDY

INSTRUCTOR NAME:- COMMANDANT AMINU IDRIS

DATE OF SUBMISSION:- 19/10/2026

Case scenario

In this training case, Harbour Operations Ltd requests a forensic review of a company-issued iPhone SE following a dispute about messages, shared media and the device's movements. As the examiner, establish the associated accounts and reconstruct relevant activity. Assess whether communications, browsing, photographs and device records support a consistent account of events.

Use the original identities and historical dates in the supplied dataset. Base your conclusions on the records examined and explain any findings that remain uncertain.

Lab Objectives

The objectives of the lab is to understand how to analyse/handle digital evidence related to an IOS operating system.

Tools and Resources Used

- Kali linux:- A penetration testing platform used for security assessments.

Kali linux tools

- Sqlitebrowser:- A light GUI editor for database

Methology

- I power on the kali linux machine
- I used the wget command to download the ios_13_4_1.zip
- I obtain some basic information such as the MD5 & SHA256 hash values of the file(ios_13_4_1.zip)
- I start analyzing the IOS file(ios_13_4_1.zip) using the different digital forensic tools.

Screenshot and Evidence

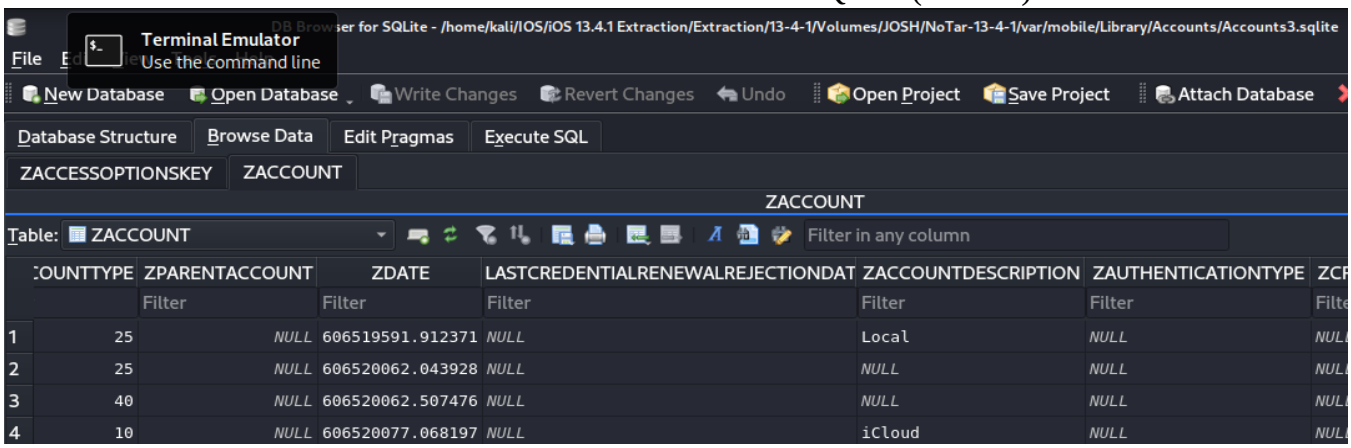
EVIDENCE PRESERVATION

SCREENSHOT SHOWING THE HASH VALUE OF THE CASE STUDY FILE (EVID-1)



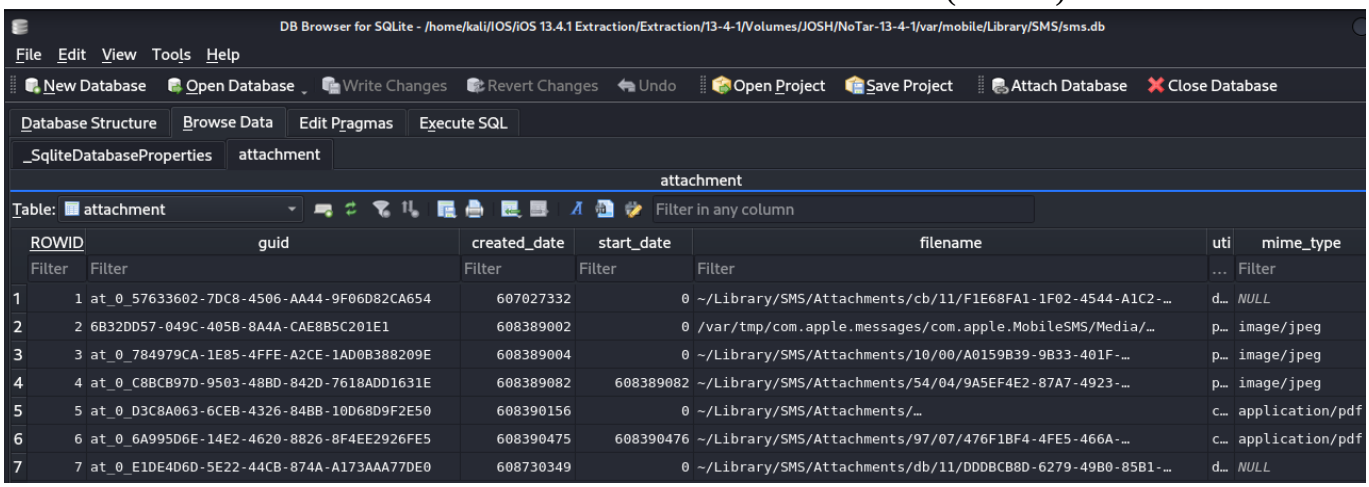
DEVICE ACCOUNTS AND SETTINGS

SCREENSHOT SHOWING THE ANALYSIS OF ACCOUNT3.SQLITE(EVID-2)



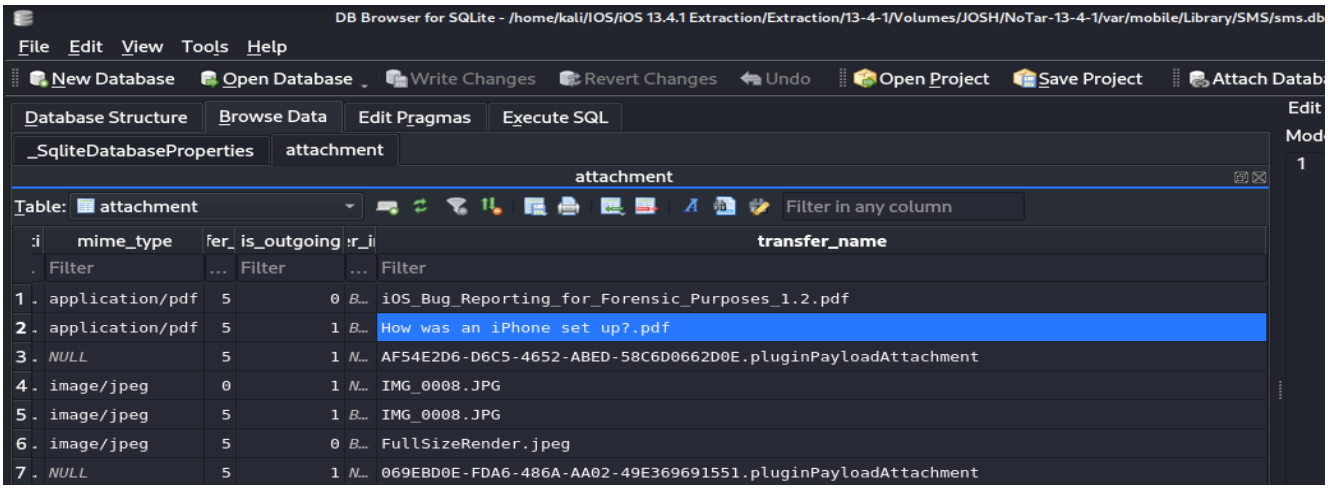
MESSAGES CONTACTS AND CALLS

SCREENSHOT SHOWING THE MESSAGE TIMESTAMP&FILE NAME(EVID-3)



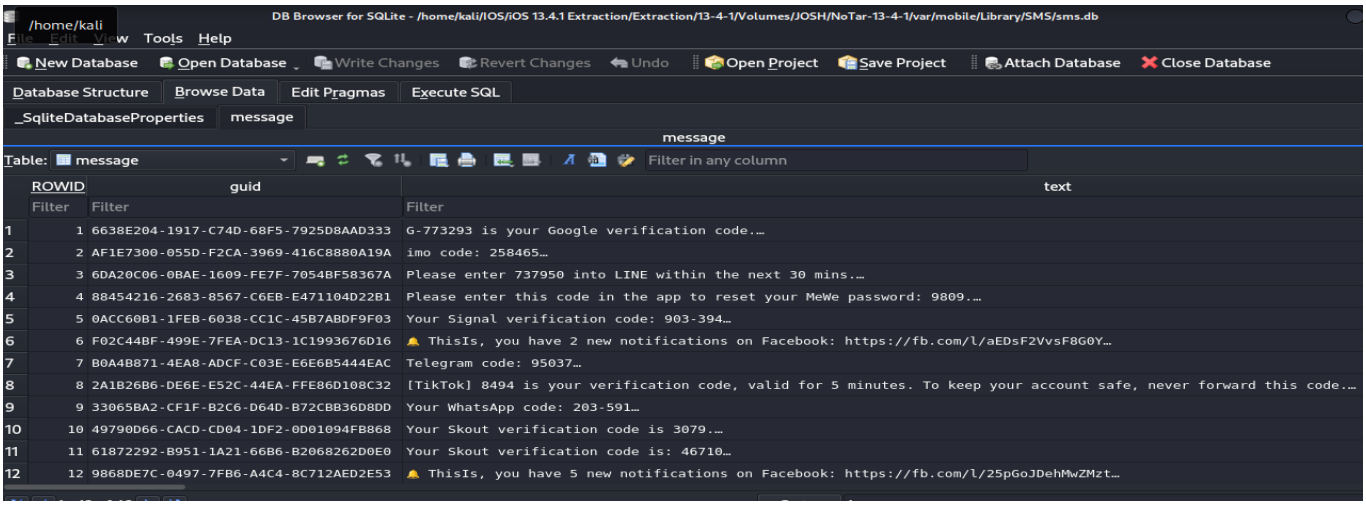
ROWID	guid	created_date	start_date	filename	uti	mime_type
1	at_0_57633602-7DC8-4506-AA44-9F06D82CA654	607027332	0	~/Library/SMS/Attachments/cb/11/F1E68FA1-1F02-4544-A1C2-…	d…	NULL
2	6B32DD57-049C-405B-8A4A-CAE8B5C201E1	608389002	0	/var/tmp/com.apple.messages/com.apple.MobileSMS/Media/…	p…	image/jpeg
3	at_0_784979CA-1E85-4FFE-A2CE-1AD0B388209E	608389004	0	~/Library/SMS/Attachments/10/00/A0159B39-9B33-401F-…	p…	image/jpeg
4	at_0_C8BCB97D-9503-48BD-842D-7618ADD1631E	608389082	608389082	~/Library/SMS/Attachments/54/04/9A5EF4E2-87A7-4923-…	p…	image/jpeg
5	at_0_D3C8A063-6CEB-4326-84BB-10D68D9F2E50	608390156	0	~/Library/SMS/Attachments/…	c…	application/pdf
6	at_0_6A995D6E-14E2-4620-8826-8F4EE2926FE5	608390475	608390476	~/Library/SMS/Attachments/97/07/476F1BF4-4FE5-466A-…	c…	application/pdf
7	at_0_E1DE4D6D-5E22-44CB-874A-A173AAA77DE0	608730349	0	~/Library/SMS/Attachments/db/11/DDDBCB8D-6279-49B0-85B1-…	d…	NULL

SCREENSHOT SHOWING THE ATTACHMENT NAME & IMAGE EXTENSION TYPE(EVID-4)

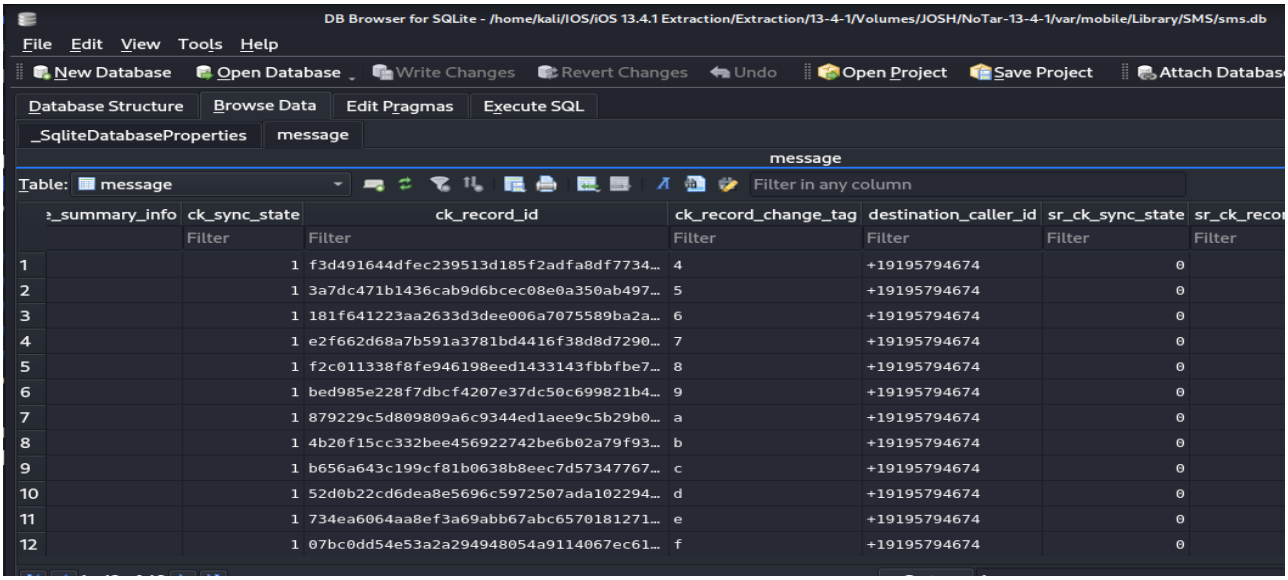


:i	mime_type	fer_	is_outgoing	:r_i	transfer_name
1	application/pdf	5	0	B…	iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf
2	application/pdf	5	1	B…	How was an iPhone set up?.pdf
3	NULL	5	1	N…	AF54E2D6-D6C5-4652-ABED-58C6D0662D0E.pluginPayloadAttachment
4	image/jpeg	0	1	N…	IMG_0008.JPG
5	image/jpeg	5	1	B…	IMG_0008.JPG
6	image/jpeg	5	0	B…	FullSizeRender.jpeg
7	NULL	5	1	N…	069EBD0E-FDA6-486A-AA02-49E369691551.pluginPayloadAttachment

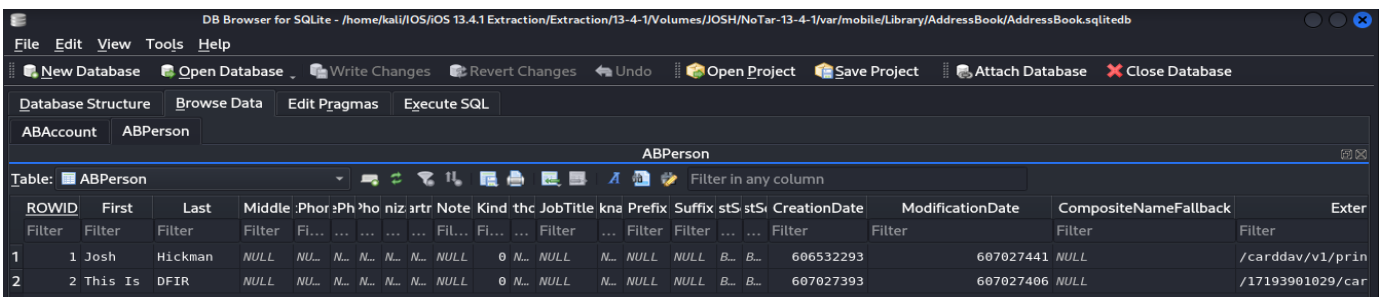
SCREENSHOT SHOWING THE MESSAGE CONTENT(EVID-5)

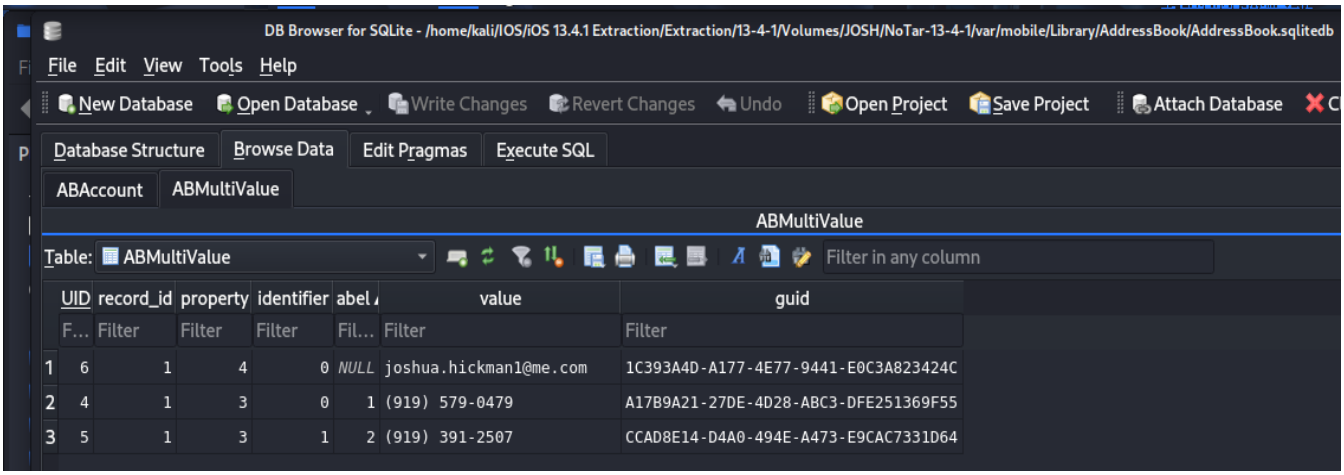


SCREENSHOT SHOWING THE MESSAGE RECIEVER(RECEIVING PHONE NUMBER) (EVID-6)



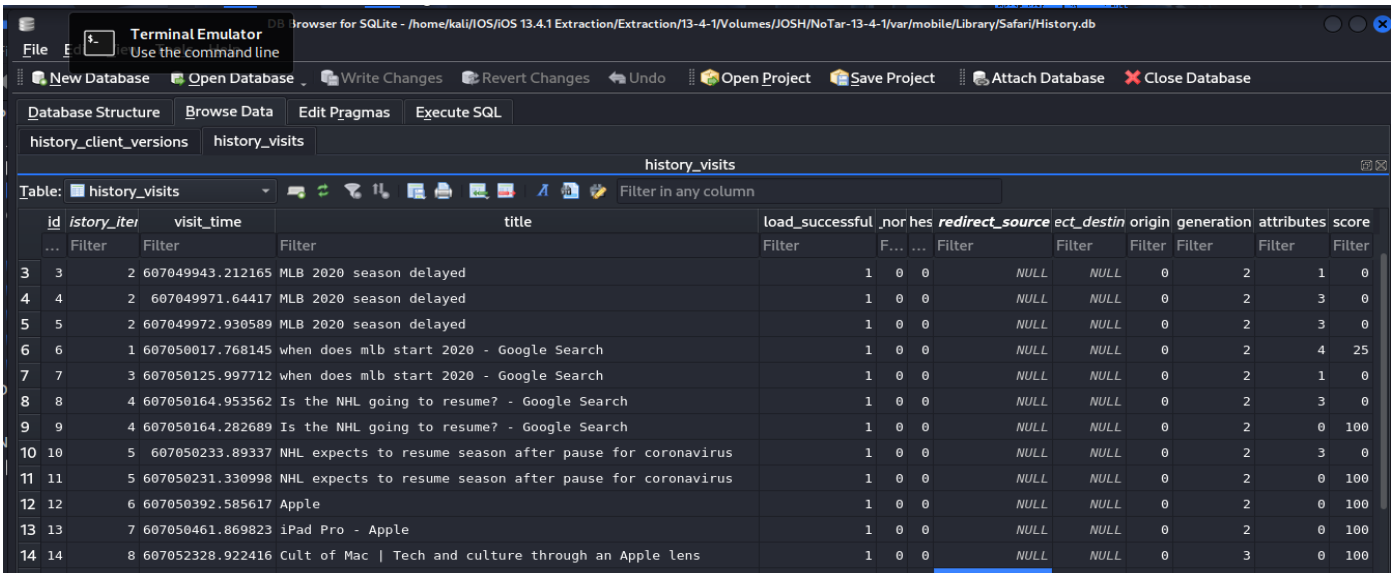
SCREENSHOT SHOWING ANALYSING THE ADDRESSBOOK.SQLITEDB(EVID-7)



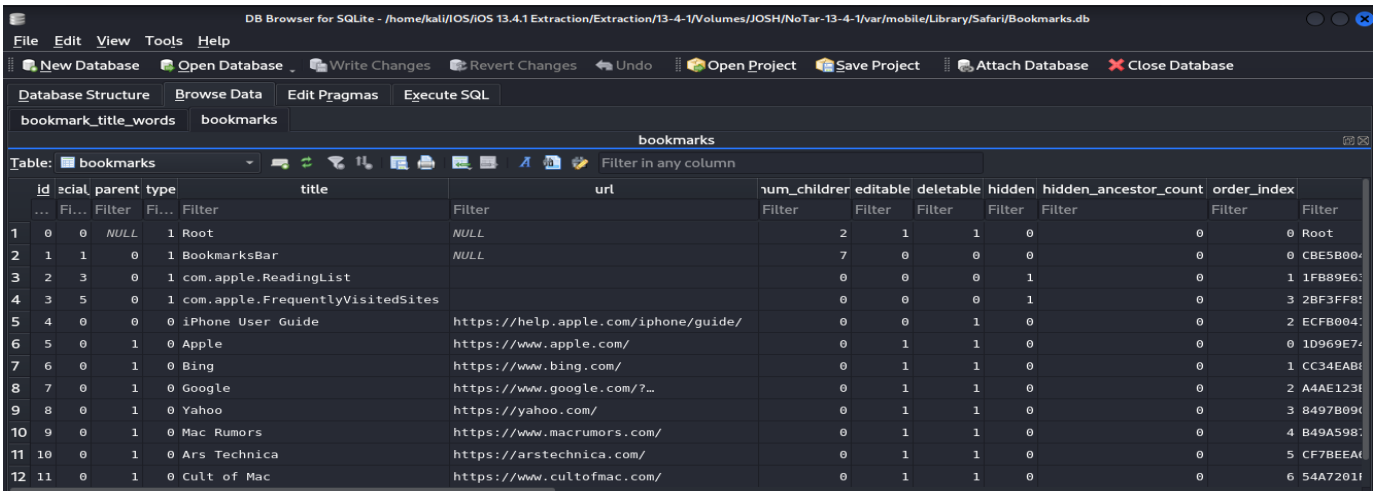


SAFARI AND CALENDAR

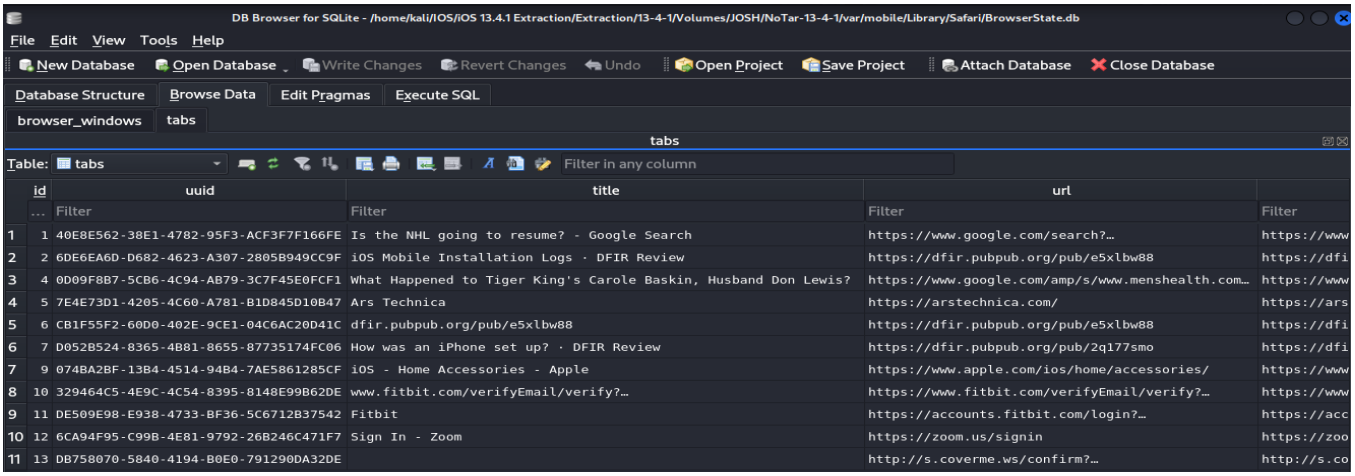
SCREENSHOT SHOWING THE SEARCH TITLE AND VISIT TIME(HISTORY.DB) (EVID-8)

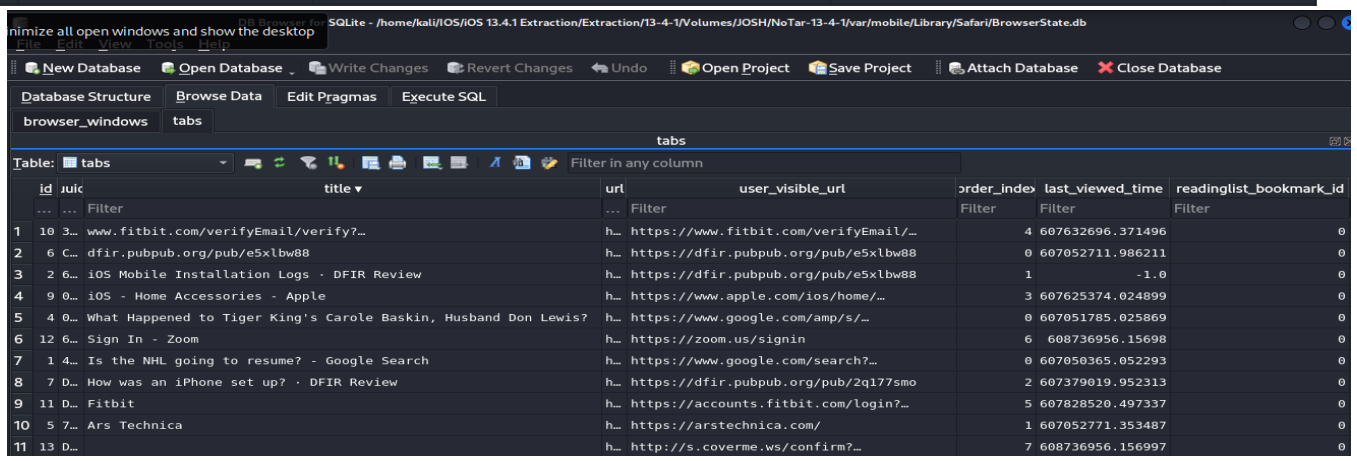


SCREENSHOT SHOWING ANALYSIS OF BOOKMARKS(BOOKMARK.DB) (EVID-9)

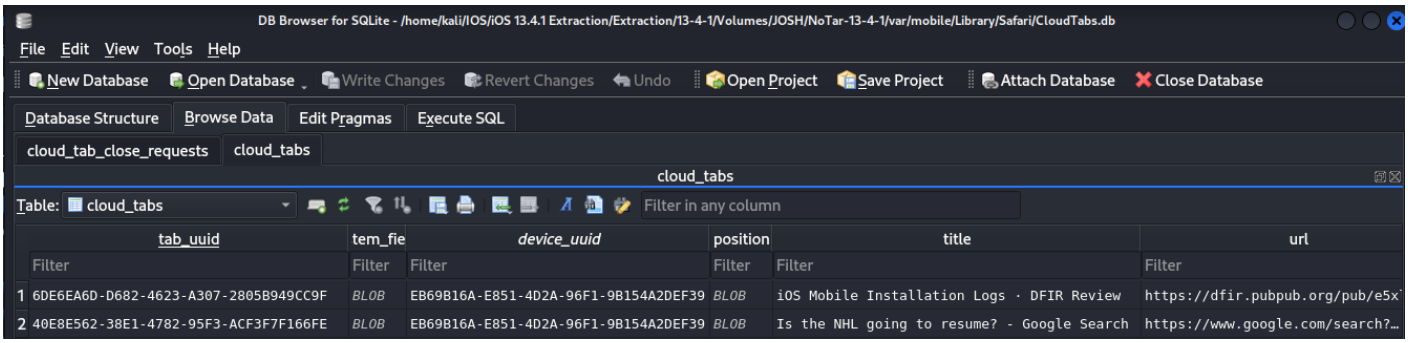


SCREENSHOT SHOWING ANALYSIS OF TABS(BROWSERSTATE.DB) (EVID-10)

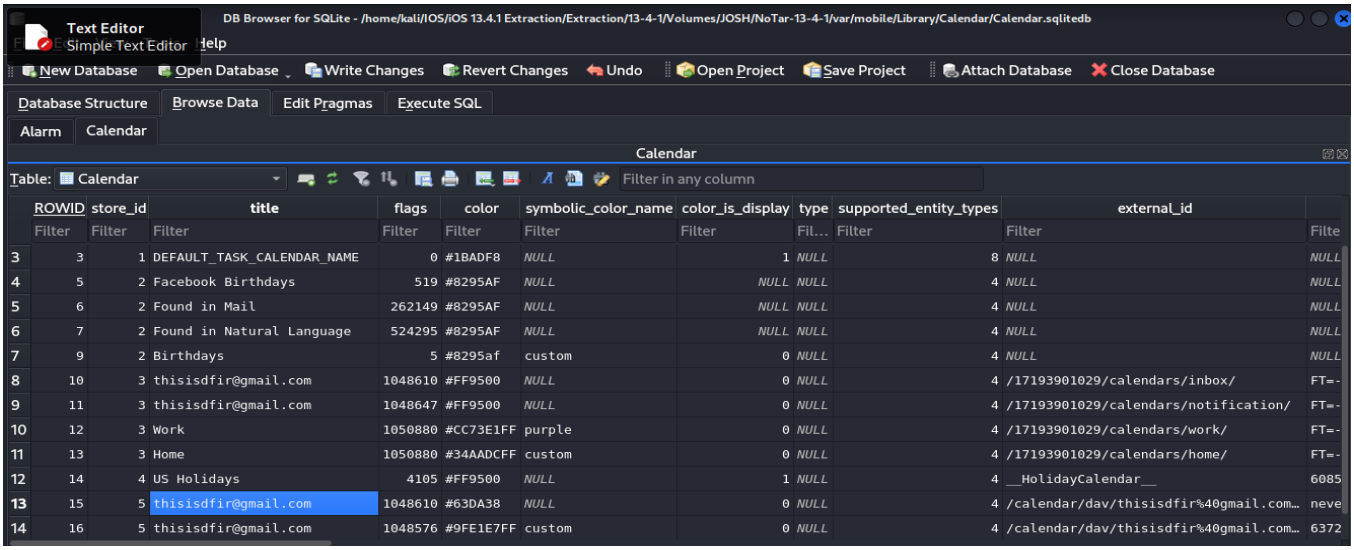




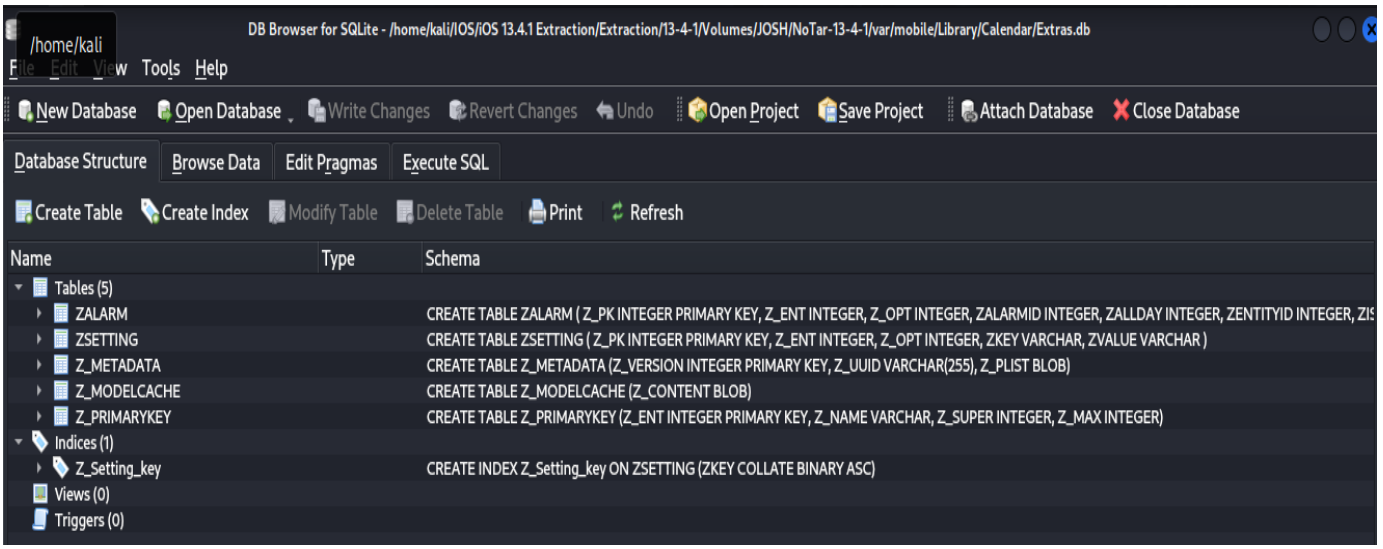
SCREENSHOT SHOWING ANALYSIS OF CLOUD_TABS(CLOUDTABS.DB) (EVID-11)



SCREENSHOT SHOWING ANALYSING CALENDER(CALENDER.SQLITEDB) (EVID-12)



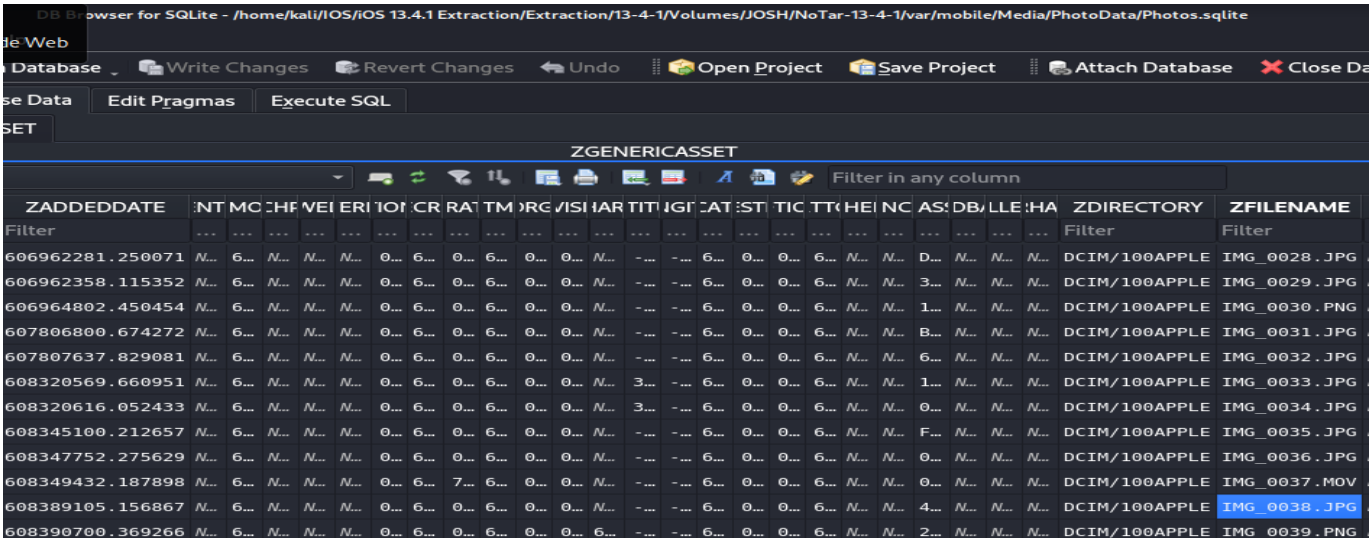
SCREENSHOT SHOWING ANALYSING(EXTRAS.DB) (EVID-13)



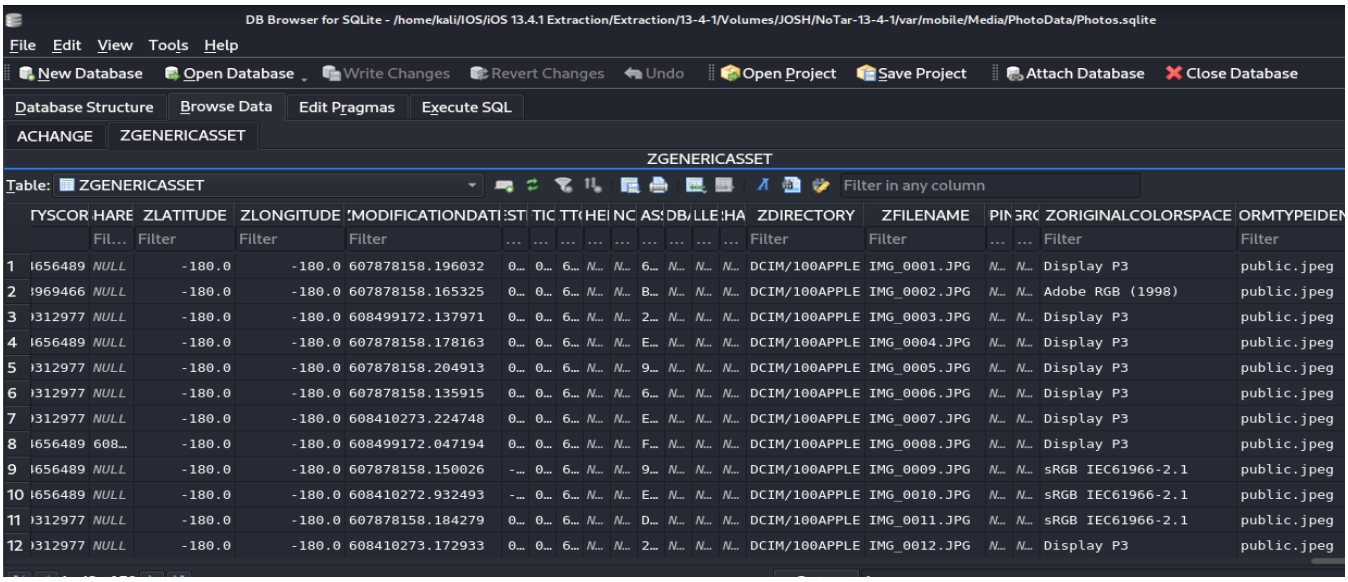
PHOTOGRAPHS AND MEDIA

SCREENSHOT SHOWING ANALYSIS OF PHOTOS.SQLITE

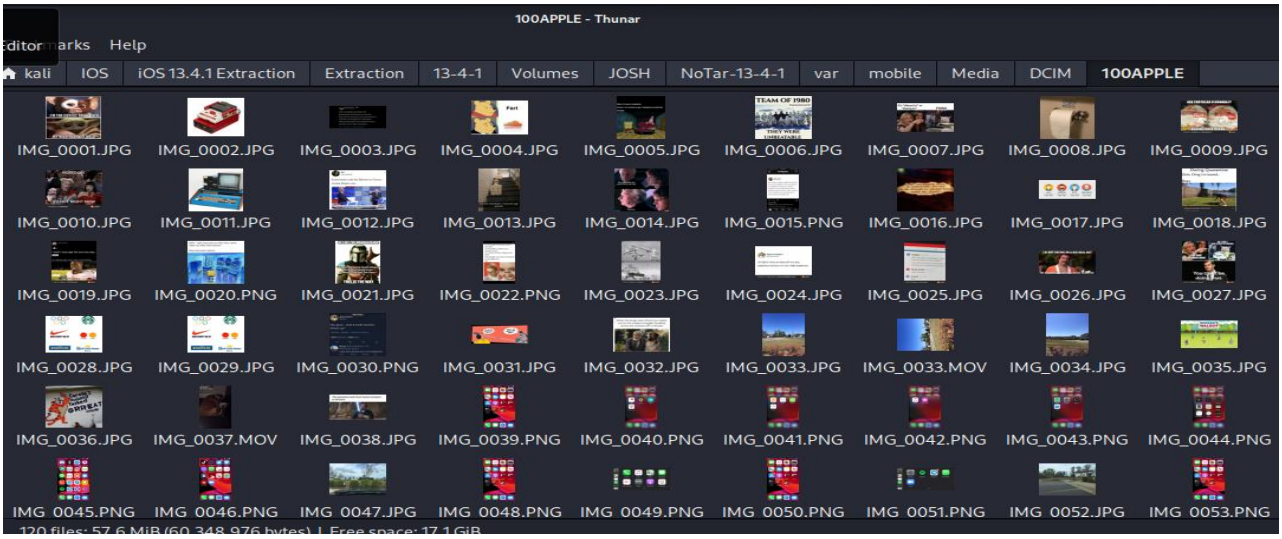
SCREENSHOT SHOWING TIMESTAMP,DIRECTORY AND FILENAME(EVID-14)



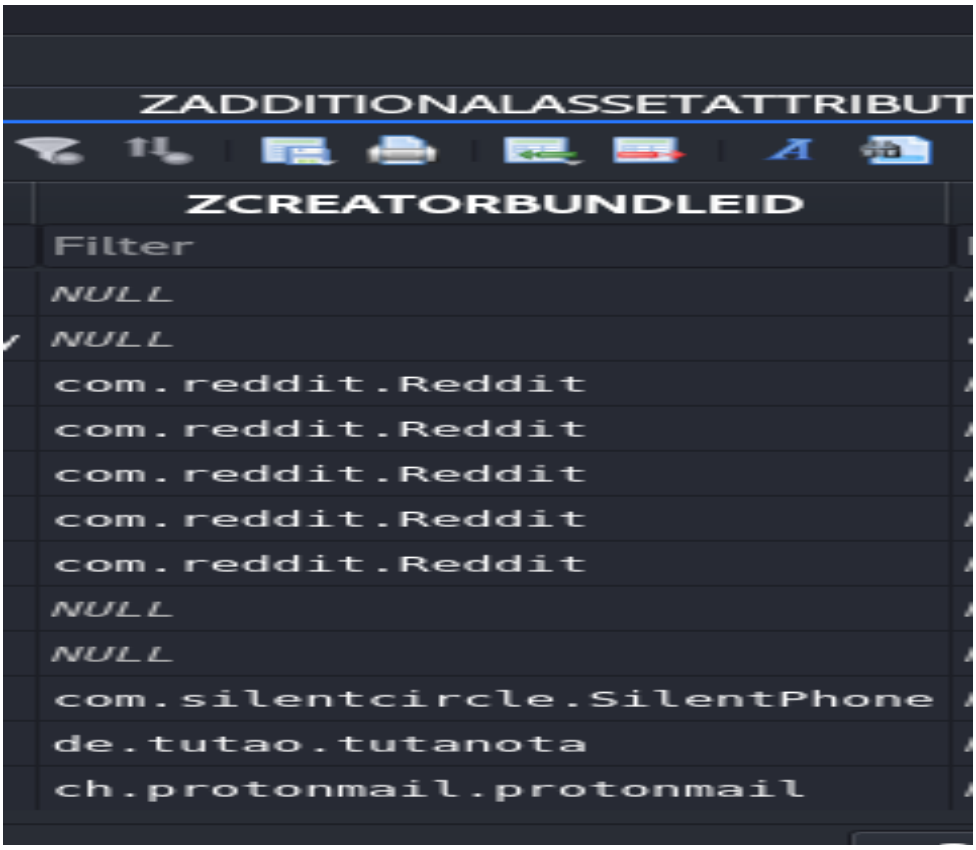
SCREENSHOT SHOWING THE LATITUDE & LONGITUDE(EVID-15)



SCREENSHOT SHOWING THE OVERALL IMAGES IN THE DCIM FOLDER(EVID-16)

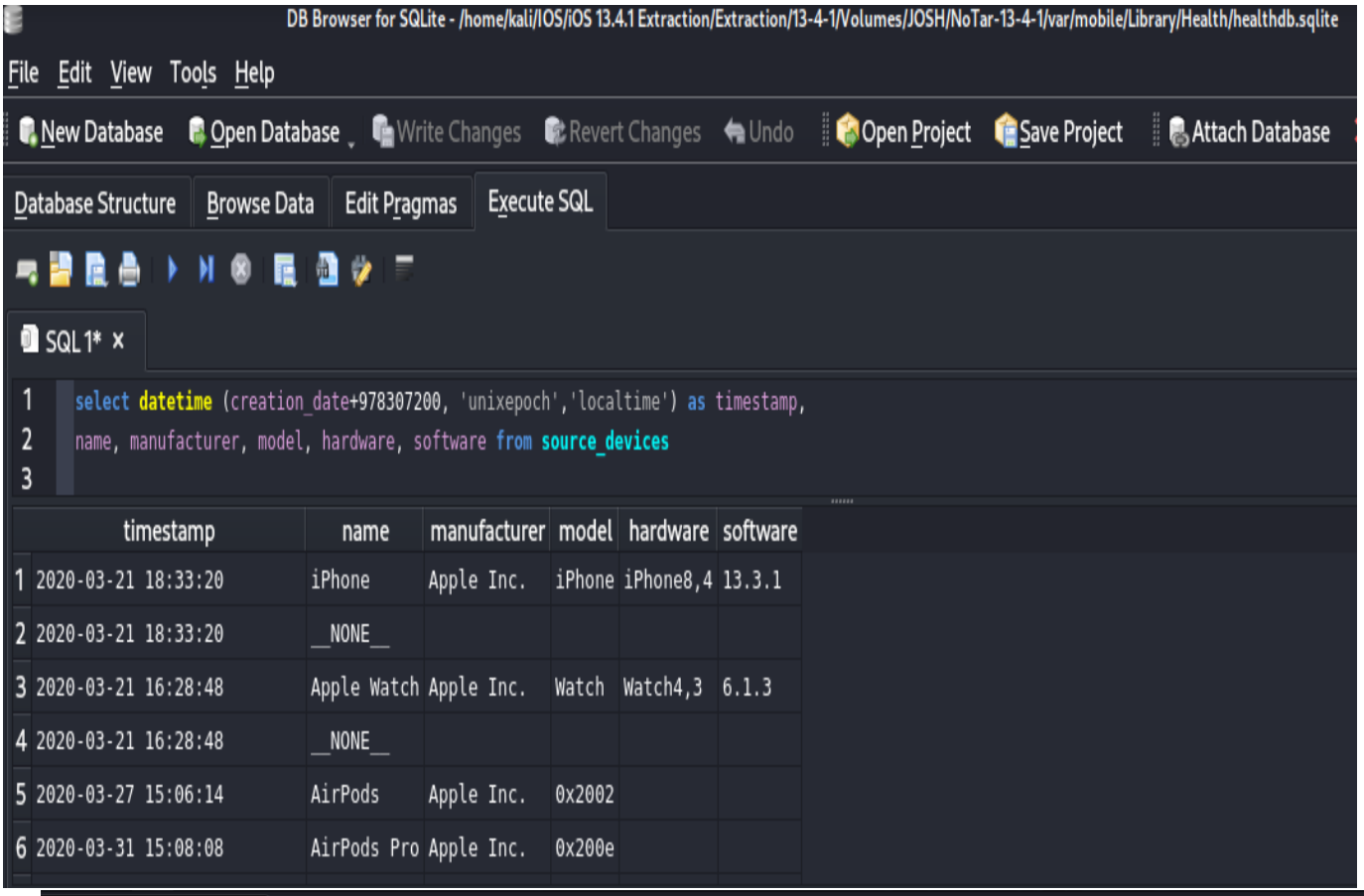


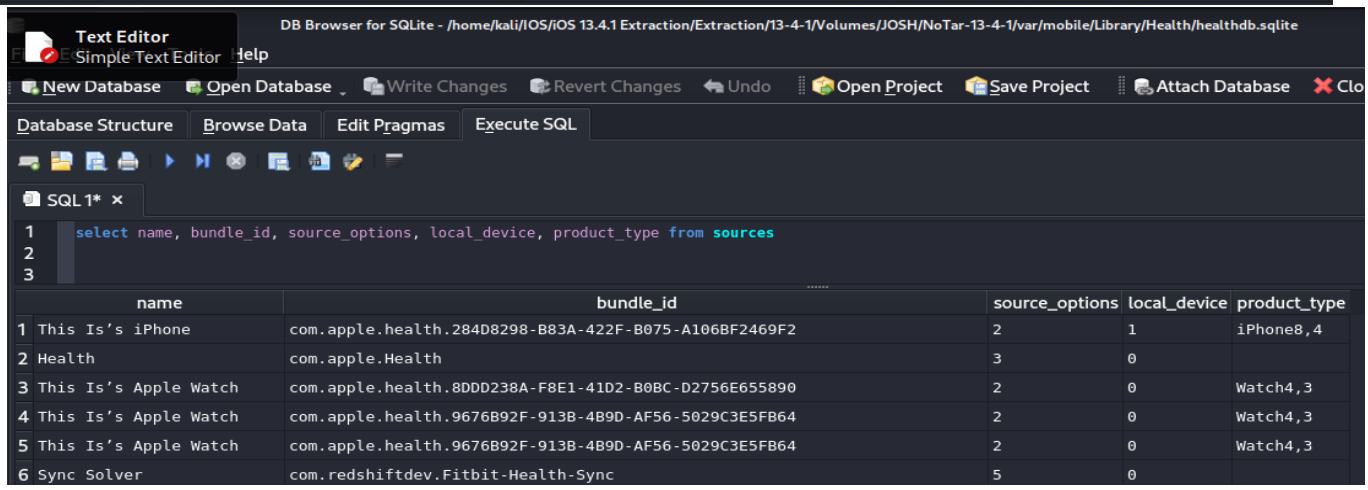
SCREENSHOT SHOWING SOME LIST OF IMAGES THAT WERE NOT TAKEN BY THE PHONE CAMERA(EVID-17)

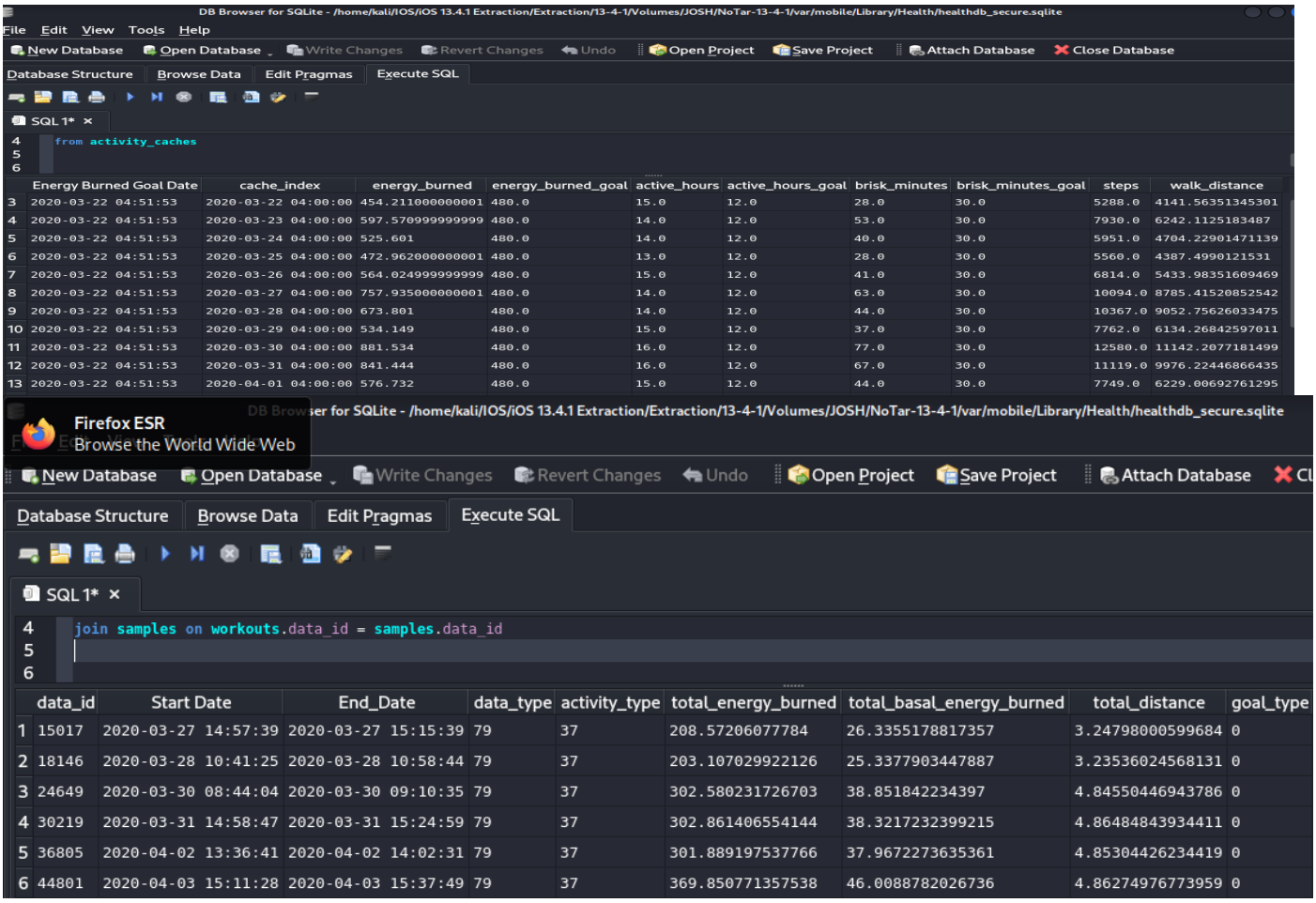


HEALTH AND LOCATION EVIDENCE

SCREENSHOT SHOWING THE ANALYSIS HEALTHDB.SQLITE & HEALTHDB_SECURE(EVID-18)


DB Browser for SQLite - /home/kali/IOS/iOS 13.4.1 Extraction/Extraction/13-4-1/Volumes/JOSH/NoTar-13-4-1/var/mobile/Library/Health/healthdb.sqlite
File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Save Project
Attach Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
SQL 1* x
1 select datetime (creation_date+978307200, 'unixepoch','localtime') as timestamp,
2 name, manufacturer, model, hardware, software from source_devices
3
timestamp name manufacturer model hardware software
1 2020-03-21 18:33:20 iPhone Apple Inc. iPhone iPhone8,4 13.3.1
2 2020-03-21 18:33:20 __NONE__
3 2020-03-21 16:28:48 Apple Watch Apple Inc. Watch Watch4,3 6.1.3
4 2020-03-21 16:28:48 __NONE__
5 2020-03-27 15:06:14 AirPods Apple Inc. 0x2002
6 2020-03-31 15:08:08 AirPods Pro Apple Inc. 0x200e


Text Editor
Simple Text Editor
DB Browser for SQLite - /home/kali/IOS/iOS 13.4.1 Extraction/Extraction/13-4-1/Volumes/JOSH/NoTar-13-4-1/var/mobile/Library/Health/healthdb.sqlite
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Save Project
Attach Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
SQL 1* x
1 select name, bundle_id, source_options, local_device, product_type from sources
2
3
name bundle_id source_options local_device product_type
1 This Is's iPhone com.apple.health.284D8298-B83A-422F-B075-A106BF2469F2 2 1 iPhone8,4
2 Health com.apple.Health 3 0
3 This Is's Apple Watch com.apple.health.8DDD238A-F8E1-41D2-B0BC-D2756E655890 2 0 Watch4,3
4 This Is's Apple Watch com.apple.health.9676B92F-913B-4B9D-AF56-5029C3E5FB64 2 0 Watch4,3
5 This Is's Apple Watch com.apple.health.9676B92F-913B-4B9D-AF56-5029C3E5FB64 2 0 Watch4,3
6 Sync Solver com.redshiftdev.Fitbit-Health-Sync 5 0



TIMELINE AND CONCLUSIONS

UTC time	Artefact field	Raw value	Epoch/unit	Evidence ID	Event description
2020-03-21 21:39:51.912 UTC	Accounts3.sqlite ZACCOUNT ZDATE	606519591.912371	Apple epoch seconds	E-02	It Stored the account row date and also account type 25
2020-03-21 21:47:42.043 UTC	Accounts3.sqlite ZACCOUNT ZDATE	606520062.043928	Apple epoch seconds	E-02	It Stored the account row date and also

					account type 25
2020-03-21 21:47:57.068 UTC	Accounts3.sqlite ZACCOUNT ZDATE	606520077.068197	Apple epoch seconds	E-02	Stored account row date and also iCloud description visible on account row
2020-03-28 00:59:03.212 UTC	History.db history_visits.visit_time	607049943.212165	Apple epoch seconds	E-8	MLB 2020 season delayed
2020-03-28 00:59:31.644 UTC	History.db history_visits.visit_time	607049971.644170	Apple epoch seconds	E-8	MLB 2020 season delayed
2020-03-28 01:00:17.768 UTC	History.db history_visits.visit_time	607050017.768145	Apple epoch seconds	E-8	A Search was made “when does mlb start 2020”
2020-03-28 01:02:44.953 UTC	History.db history_visits.visit_time	607050164.953562	Apple epoch seconds	E-8	A Search was made “Is the NHL going to resume?”
2020-03-28 01:03:53.893 UTC	History.db history_visits.visit_time	607050233.893370	Apple epoch seconds	E-08	NHL expects to resume season after pause for coronavirus
2020-03-28 01:38:48.922 UTC	History.db history_visits.visit_time	607052328.922416	Apple epoch seconds	E-08	iPad Pro - Apple
2020-04-03 18:51:36.371 UTC	BrowserState.db tabs.last_viewed_time	607632696.371496	Apple epoch seconds	E-10	This is the recorded timestamp visible on

					the attachment
2020-04-12 12:56:42 UTC	sms.db attachment.created_date	608389002	Apple epoch seconds	E-03	This is the recorded timestamp visible on the attachment
2020-04-12 12:56:44 UTC	sms.db attachment.created_date	608389004	Apple epoch seconds	E-03	Attachment record timestamp

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Analysis and Discussion

RESULT OBTAIN FROM EVIDENCE PRESERVATION

Item	**Value shown in supplied screenshot**
Source archive	ios_13_4_1.zip
Size	9,629,286,492 bytes
MD5	4e094801ba1e3b938b2c0cc18dded1d
SHA-256	761854a669c38da6acda908afe77763d54ccb83044ff123350890d408c154551

RESULT OBTAIN FROM SAFARI AND CALENDAR

Visible title	**Raw visit_time (Apple epoch seconds)**	**UTC conversion**
when does mlb start 2020 - Google Search	607050125.997712	2020-03-28 01:02:05.997 UTC
Is the NHL going to resume? - Google Search	607050164.953562	2020-03-28 01:02:44.953 UTC
NHL expects to resume season after pause for coronavirus	607050233.893370	2020-03-28 01:03:53.893 UTC
Apple	607050392.585617	2020-03-28 01:06:32.585 UTC
iPad Pro - Apple	607052328.922416	2020-03-28 01:38:48.922 UTC

RESULT OBTAIN FROM PHOTOGRAPHS AND MEDIA

Filename	Raw ZADDEDDATE	UTC conversion	Interpretation
IMG_0028.JPG	606962281.250071	2020-03-27 00:38:01.250 UTC	The ZADDEDDATE represent the timestamp
IMG_0029.JPG	606962358.115352	2020-03-27 00:39:18.115 UTC	The ZADDEDDATE represent the timestamp
IMG_0030.PNG	606964802.450454	2020-03-27 01:20:02.450 UTC	The ZADDEDDATE represent the timestamp

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Conclusion

In conclusion this lab has help me with the basic concept,skills and understanding that I may require my handling or dealing with a digital evidence that utilizes an ios operating system.

Reference

➢ICDFA course Material (SBT-DF205(MOBILE AND IOT FORENSICS) by COMMANDANT AMINU IDRIS
